



Faculty of Engineering and Technology
Computer Science Department
Data Base Systems COMP333

Assignment No.1

Prepared by:

Student Name: **Ayah Abdalmuti**

Student ID: **1212074**

Student Name: **Rama Shaheen**

Student ID: **1211153**

Supervised by:
Dr. Bashar Tahayna

Question#1

Name the main entities you would include in the ERD ?

- **Flight**
- **AirPalne**
- **Ticket**
- **Passenger**
- **Employee**

Question#2

What is the key attribute that can uniquely identify each Flight instance?

- **Flight_Number**

Question#3

Can a Flight exist in the system without an assigned Airplane? Explain why?

Yes , a flight can exist in the system without an assigned airplane, because the flight might be scheduled in the system but not yet assigned an airplane.

Question#4

List the possible relationships between the Employee and Flight entities?

- **Pilot assigned to Flight:** Each flight can have one or more pilot assigned to it, while a pilot can be assigned to multiple flights over time.
- **Crew assigned to Flight:** Each flight can have one or more crew assigned to it, while a crew can be assigned to multiple flights.
- **Ground Operations Staff assigned to Flight:** Ground members can be assigned to specific flights for tasks like loading baggage, preparing the aircraft, or assisting passengers.

Question#5

Between Flight and Airplane, which entity is the owner of the relationship? Briefly justify?

Flight is the owner of the realationship ,because the foreign key of Airplane (AirPlane_ID) is exist in the Flight table.

Question#6

The Ticket and Flight entities are related through which kind of multi-valued dependency? a) One to one b) One to many c) Many to one d) Many to many

Many Ticket to one flight , c.(Many to One).

Question#7

Name the attributes you would include for Passenger along with data types for each?

- **Name : VarChar**
- **PassengerID : INT**
- **ContactInfo : VarChar**
- **FrequentFlyerDetails : VarChar**

Question#8

Should Ticket be modeled as a weak entity? Why or why not?

Yes The Ticket is a weak entity , because the ticket exist only when the flight and passenger exist.

Question#9

What is the cardinality between the Flight and Airplane entities?

- a) One-to-one b) One-to-many c) Many-to-one d) Many-to-many

Many Flight for One Airplane , c.(Many to one)

Question#10

Passenger bookings involve a many-to-many relationship between which two entities? a) Passenger and Ticket b) Airplane and Flight c) Passenger and Flight d) Ticket and Flight

C.(Passenger and Flight)

Question#11

The Pilot entity refers to staff piloting flights. What is its primary key and list 2 other attributes?

- **Piolt_ID : PrimrayKey**
- **Piolt_Name**
- **Piolt_NumberofFlight**

Question#12

Will you recommend using a weak entity construct in your ERD? If yes, which one and why?

Ticket , because the ticket exist only when the flight and passenger exist.

Question#13

What type of relationship exists between Ticket and Flight entities? Define constraints to enforce referential integrity.

Many Ticket to One Flight (Many to One) , because the Foreign key like (Flight_ID) Is exist In ticket table.

